

Question number 1.

Your answer was E. Correct. 

Give the differential equation that has $y = C_1 + C_2x^4 - 3x$ as its general solution.

- A $x y' - 3y = 9$
- B $x y'' + 3y' = -9$
- C $x y'' + 5y' = 9$
- D $x y'' - 3y' = -15$
- E $x y'' - 3y' = 9$
- F None of the above.

Question number 2.

Your answer was C. Correct. 

Find the general solution of $(x^2y + 3y)y' = 2xy^2 + 8x$.

- A $y^2 + 4 = (x^2 + 3)^2 + C$
- B $y^2 = C(x^2 + 3) - 4$
- C $y^2 = C(x^2 + 3)^2 - 4$

D $y^2 = (x^2 + 3)^2 + C$

E $y^2 + 4 = (x^2 + 3) + C$

F None of the above.

Question number 3.

Your answer was E. Correct. 

Find the general solution of $xy' - 4y = x^2 - 3$.

A $y = \frac{1}{2}x^2 + \frac{3}{4}x^4 + Cx^4$

B $y = -\frac{1}{2}x^{-2} + \frac{3}{4}x + Cx^4$

C $y = -\frac{1}{2}x^2 + \frac{3}{4} + Cx^{-4}$

D $y = -\frac{1}{8}x^2 - 3x^{-4} + C$

E $y = -\frac{1}{2}x^2 + \frac{3}{4} + Cx^4$

F None of the above.

Question number 4.

Your answer was D. Correct. 

A certain radioactive material is decaying at a rate proportional to the amount present. A sample of 100 grams of the material was present initially and after 4 hours the sample lost 10% of its mass. Find the half-life of the material.

A $T = \frac{-\ln(2)}{\ln\left(\frac{9}{10}\right)}$

B $T = \frac{-\ln(2)}{\ln(9)}$

C $T = \frac{-4\ln(2)}{\ln\left(\frac{10}{9}\right)}$

D $T = \frac{-4\ln(2)}{\ln\left(\frac{9}{10}\right)}$

E $T = \frac{4\ln(2)}{\ln\left(\frac{9}{10}\right)}$

F None of the above.

Question number 5.

Your answer was B. Correct. 

Give the general solution of $y'' + 6y' + 34y = 0$.

A $y = C_1 e^{3x} \cos(5x) + C_2 e^{3x} \sin(5x)$

B $y = C_1 e^{-3x} \cos(5x) + C_2 e^{-3x} \sin(5x)$

C $y = C_1 e^{-3x} + C_2 e^{5x}$

D $y = C_1 e^{3x} + C_2 e^{-5x}$

E $y = C_1 e^{5x} \cos(3x) + C_2 e^{5x} \sin(3x)$

F None of the above.

Question number 6.

Your answer was B. Correct. 

A fundamental set of solutions of $y'' + 8y' + 16y = 0$ is:

A $\{e^{-4x}, e^{-4x}\}$

B $\{e^{-4x}, xe^{-4x}\}$

C $\{e^{-4x}, xe^{4x}\}$

D $\{e^{4x}, xe^{4x}\}$

E $\{e^{-4x}, e^{4x}\}$

F None of the above.

Question number 7.

Your answer was D. Wrong. 

The function $y(x) = -5e^x - 3e^{-3x}$ is a solution of a second order linear homogeneous differential equation with constant coefficients. The equation is:

A $y'' - 2y' - 2y = 0$

B $y'' + 2y' - 3y = 0$

C $y'' + 4y' + 3y = 0$

D $y'' - 2y' - 3y = 0$

E $y'' - 4y' - 3y = 0$

F None of the above.

Question number 8.

Your answer was A. Correct. 

This is a written question, worth 10 points. DO NOT place the problem code on the answer sheet. A proctor will fill this out after exam submission. Show all steps (work) on your answer sheet for full credit.
Problem Code: 851

(a) In each of (1) and (2), determine whether the given equation is linear, separable, Bernoulli, homogeneous, or none of these.

(1) $(2y + x^2y)y' = 2xy + xy^3$

(2) $xy' + x^2y = \frac{3x^3}{y}$

(b) Find the general solution of $xy' = 4x^2\sqrt{y} - 4y$.

- A I have placed my work and my answer on my answer sheet.
- B I want to have points deducted from my test for not working this problem.

Question number 9.

Your answer was A. Correct. 

This is a written question, worth 10 points. DO NOT place the problem code on the answer sheet. A proctor will fill this out after exam submission. Show all steps (work) on your answer sheet for full credit.
Problem Code: 951

(a) Find the differential equation for the family of curves $y^4 = Cx^3 - 3x$.

(b) $y' = \frac{3y^4 + 6}{4xy^3}$ is the differential equation for a family of curves $F(x, y, C) = 0$. Find the family of orthogonal trajectories.

- A I have placed my work and my answer on my answer sheet.
- B I want to have points deducted from my test for not working this problem.

Question number 10.

Your answer was A. Correct. 

This is a written question, worth 10 points. DO NOT place the problem code on the answer sheet. A proctor will fill this out after exam submission. Show all steps (work) on your answer sheet for full credit.
Problem Code: 1044

(a) The differential equation $y'' + \frac{3}{x}y' - \frac{8}{x^2}y = 0$ has two solutions of the form $y = x^r$. Find two independent solutions y_1, y_2 of the equation and calculate their Wronskian.

(b) Consider the initial-value problem

$$y'' + p(x)y' + q(x)y = 0, y(1) = 2, y'(1) = 4.$$

The functions $u_1(x) = 2x^3$, $u_2(x) = 3x^4 - x^3$, $u_3(x) = -x^4$ are solutions of the differential equation. Give the general solution of the equation and find the solution of the initial-value problem.

- A I have placed my work and my answer on my answer sheet.
- B I want to have points deducted from my test for not working this problem.

Question number 11.

Your answer was A. Correct. 

This is a BONUS written question, worth 2 points. DO NOT place the problem code on the answer sheet. A proctor will fill this out after exam submission. Show all steps (work) on your answer sheet for full credit.

Problem Code: 1155

A disease is infecting a flock of 100 sheep living on a small, remote ranch. Let $S(t)$ be the number of sick sheep t days after the outbreak. Suppose that 5 sheep had the disease initially, and suppose that the disease is spreading at a rate proportional the product of the time elapsed and the number of sheep that do not have the disease.

Give the mathematical model (initial-value problem) for S .

- A I have placed my work and my answer on my answer sheet.
 - B I want to have points deducted from my test for not working this problem.
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